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Improved velocity-Verlet algorithm for the discrete element method

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ABSTRACT

The Discrete Element Method is widely employed for simulating granular flows, but conventional integration techniques may produce unphysical results for simulations with static friction when particle size ratios exceed $\mathscr{R} \approx 3$. These inaccuracies arise under certain circumstances because some variables in the velocity-Verlet algorithm are calculated at the half-timestep, while others are computed at the full timestep. To correct this, we develop an improved velocity-Verlet integration algorithm to ensure physically accurate outcomes up to the largest size ratios examined ($\mathscr{R} = 100$). The implementation of this improved `synchronized_verlet` integration method within the LAMMPS framework is detailed, and its effectiveness is validated through a simple three-particle test case and a more general example of granular flow in mixtures with large size-ratios, for which we provide general guidelines for selecting simulation parameters and accurately modeling inelasticity in large particle size-ratio simulations.

1. Introduction

Simulation codes and the models and algorithms buried deep within them can sometimes generate unphysical or conflicting results. A famous recent case is the seven-year-long controversy over supercooled water, where slight differences in molecular dynamics codes resulted in strikingly different results [1]. Here we describe and remedy a similar subtle problem in several open-source Discrete Element Method (DEM) codes used for analyzing granular flows.

Granular materials, which include particles ranging in size from micrometers to meters, are the second most used substances in industry after water [2] and are encountered in a variety of industrial sectors such as pharmaceuticals [3], construction materials [4], food processing [5] and chemical processing [6]. Granular materials are also integral to natural and geological phenomena like avalanches, landslides, and asteroid impacts [7–9]. In many practical applications involving granular materials, particle size often varies widely; for example, uncrushed rock in mining operations is thousands of times larger than the resulting crushed fine powder [10], soil particles may range from coarse sand to fine clay [11], and debris from avalanches and asteroids can vary greatly in scale [12]. Likewise, many processes in the chemical and pharmaceutical industries involve particles with sizes spanning two or more orders of magnitude [13,14]. Accurately accounting for these size variations is crucial for research related to these applications.

One of the most commonly used numerical methods for simulating granular flow applications is the Discrete Element Method (DEM) [15,16]. In DEM, which considers 'soft' particles, local particle deformations are assumed to be small and are expressed as overlaps between adjacent contacting particles. Overlaps are used to calculate the contact forces, which determine the subsequent changes in velocity and configuration of the particles.

Many open-source DEM codes exist for simulating granular systems. Among the most widely used [17] are LAMMPS [18], LIGGGHTS [19], YADE [20], MercuryDPM [21], MUSEN [22], GranOO [23] and MFIX [24]. These solvers have been adopted to simulate granular flows in various applications such as hoppers [25,26], mills [27], mixers [28], drums [29], and granular compaction [30], as well as in geophysical and astrophysical applications such as avalanches [31], volcanoes [32] and earthquakes [33]. Most of these specific simulation examples use relatively small ranges of particle sizes, with typical large to small particle size ratios $\mathscr{R} < 3$, primarily because of the computational burden associated with larger size ratios. This limited size ratio range stands in stark contrast to the wide range of particle sizes observed in real-world granular systems.

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One of the primary challenges in DEM simulations involving polydisperse systems is the rapid increase in computational resources needed as the size ratio increases. Berger et al. [34] highlight the difficulties of simulating highly polydisperse systems, including the need for shorter timesteps to accommodate interactions between particles with large size differences, and the increased complexity in neighbor detection and parallelization. However, over the past decade, advances in computational power and algorithms have facilitated more robust simulations that make possible considering physically relevant systems with size ratios $\mathcal{R} > 3$. For instance, Remond [35] successfully simulated static granular beds with $\mathcal{R}$ up to 12.5; Gao et al. [36,37] considered flowing particles in sheared beds with $\mathcal{R}$ reaching 10; Lominé and Oger [38] leveraged linked list algorithms for improved neighbor identification, simulating fine particle percolation through static particles with $\mathcal{R}$ up to 20.

A breakthrough for simulating large size ratios was achieved with Ogarko and Luding's [39] implementation of the hierarchical grid algorithm for neighbor detection, which can be effectively parallelized and showed a 220-fold speed increase over the classical linked cell method for $\mathcal{R} = 50$. The integration of this algorithm into open-source platforms such as LAMMPS [40], MercuryDPM [21] and MUSEN [22] has propelled further studies, enabling simulations with extremely large size ratios of up to $\mathcal{R} = 200$ [41,42]. These advances underscore the growing interest and capability in exploring and simulating granular systems with large size ratios.

When we started investigating fine particle percolation in static beds, we aimed to extend existing work [35–38] to size ratios as high as $\mathcal{R} = 50$. During this process, we discovered that DEM modeling approaches developed for low particle size ratios produce unphysical results for large size ratios. For example, at $\mathcal{R} = 20$, most fine particles (fines) percolating through a static bed are expected to freely pass through the interstices between the larger bed particles. However, we observe that many fines become trapped between two larger particles, oscillating along circular paths around the line connecting the two large particles. Further investigation shows that this unphysical pendular motion occurs at size ratios as small as $\mathcal{R} = 3$. Thus, traditional DEM modeling approaches, though effective for small size ratios, $\mathcal{R} < 3$, can result in unphysical and erroneous outcomes at larger size ratios under certain circumstances such as in static beds with low fine particle concentrations. In this paper, we address these issues using the open-source package LAMMPS, which is chosen for its wide utilization, scalability and integration of the hierarchical grid algorithm. However, similar unphysical behavior also occurs with other standard DEM codes.

The primary focus of this paper is to delineate the nuances of the integration scheme used in DEM simulations, particularly focusing on its impact on the accuracy of the tangential static frictional force. Through a simple three-particle problem, we demonstrate how errors related to the velocity-Verlet integration scheme in current versions of widely-used open-source codes can lead to unphysical contact behavior, especially in systems with large size ratios. To address this problem, we propose an alternative implementation of the velocity-Verlet scheme that corrects these inaccuracies, and we validate this approach by comparing it with analytical results. Furthermore, we emphasize the importance of selecting appropriate contact stiffness values and timesteps for simulations involving large size ratios, providing guidelines for these selections. We also discuss the limitations of existing damping models in LAMMPS for accurately modeling inelasticity in large size ratio contacts and propose an alternative. Several of these enhancements have been integrated into the latest developer version of LAMMPS, offering the community improved simulation accuracy for large size ratio systems. While the velocity-Verlet scheme is used as an example due to its implementation in most open-source codes including LAMMPS, LIGGGHTS, MercuryDPM, and GRANOO, the issues described here are relevant to other integration schemes as well.

This paper is organized as follows. Section 2 introduces a simple three-particle test case and highlights the unphysical behavior caused by the standard velocity-Verlet integration scheme. Section 3 explores the root cause of this issue in the velocity-Verlet method, and Section 4 presents an improved version of the method that corrects these inaccuracies. In Section 5, the results from the improved scheme are compared with the standard approach using a particle percolation example, along with a discussion on selecting optimal spring stiffness, timesteps, and damping models for systems with large $\mathcal{R}$.

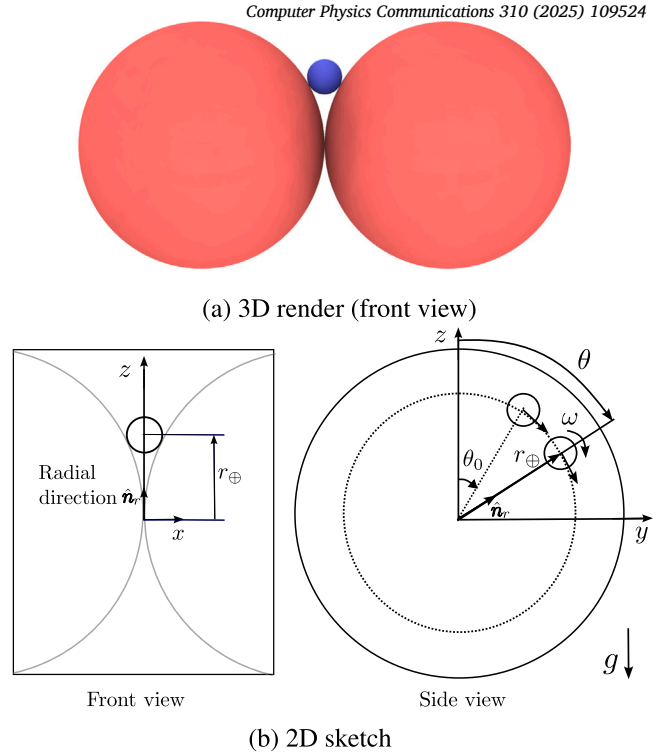

Fig. 1. (a) 3D front view and (b) 2D sketch showing the front and side view of a spherical fine particle with diameter $2r$ released from rest at initial angle θ_0 and rolling down the valley between two larger spherical particles with diameters $2R$ having their contact axis in the horizontal plane for size ratio $\mathcal{R} = 7$. The distance of the fine particle from the line connecting the centers of the two large particles is $r_\oplus$ and the unit vector in the radial direction is $\hat{\boldsymbol{n}}_r$.

2. Fine particle rolling between a large-particle pair

To illustrate and understand the issue associated with the standard velocity-Verlet approach when simulating systems with large size ratios, we consider a simple three-body problem of a spherical fine particle rolling down the valley between a contacting pair of larger stationary spherical particles, see Fig. 1. The fine particle, which is initially tangent to both large particles, is released from rest at an initial angle θ_0 with respect to vertical and moves along the valley between the two large particles under the influence of gravity, which acts vertically downward.

An analytical solution to this problem, detailed in Appendix A, is obtained by assuming that when a rigid fine particle is released at angle θ_0, it initially undergoes pure rolling motion while contacting both large particles. Once the tangential friction force F_t reaches the sliding limit μF_n, where μ is the dynamic coefficient of friction and F_n is the normal force, the particle begins to slide in addition to rolling, eventually losing contact with the two large particles and falling away. Consider, as an example, the time evolution of the position, θ, and angular velocity, ω, of the fine particle having diameter $2r$ in Fig. 1, with large particle diameters $2R = 4$ mm, particle density $\rho = 2500$ kg/m^3, $\mu = 0.6$, $g = 9.81$ m/s^2, and a large-to-fine particle size ratio of $\mathcal{R} = R/r = 7$, shown as solid curves in Fig. 2. For all three values of θ_0, the fine particle separates from the large particles at $\theta \approx 61°$, indicated by the $\times$ in Fig. 2(a). At this instant, the net radial force equals zero as the centrifugal accel-

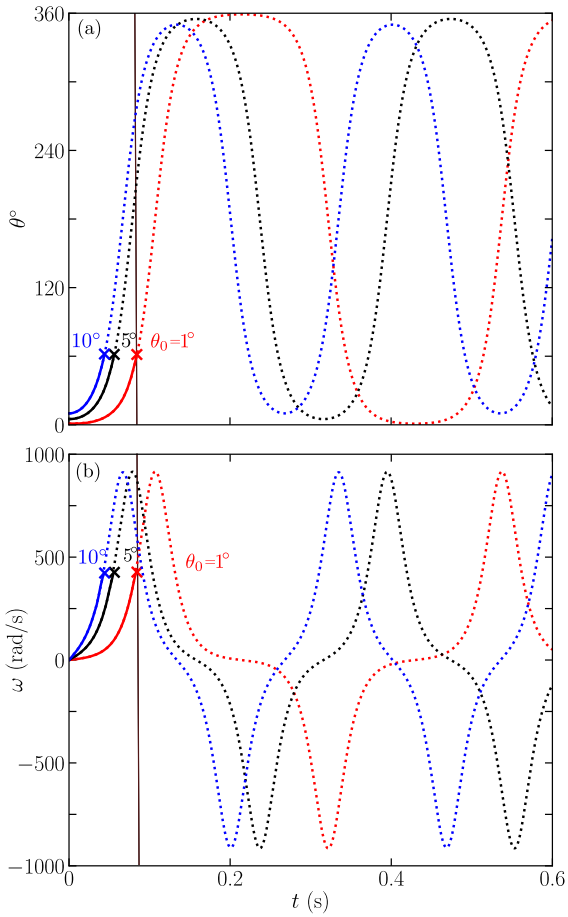

Fig. 2. Comparison of DEM predictions (dotted curves) of (a) angular position, θ and (b) angular velocity, ω, with the analytic solutions (solid curves) for different initial fine particle positions, θ_0, for the three-particle case shown in Figs. 1 and 15 with $\mathscr{R} = 7$ and $\mu = 0.6$. Note that analytic solution for motion during contact ends at $\theta \approx 61°$ when the fine particle separates from the large particles (denoted as ×) with the vertical line representing the instant when the fine particle separates from the large particles for $\theta_0 = 1°$.

eration exceeds the radial component of gravitational acceleration. For small θ_0, it takes longer to build up the angular velocity, ω, necessary for separation than for large θ_0, but the value for ω when the fine particle detaches is similar in all three cases, shown as the solid curves in Fig. 2(b).

For comparison, DEM simulations of this problem are performed in LAMMPS using the `linear_history` model with material properties of soda-lime glass [43]: Young's modulus $E = 72$ GPa, Poisson ratio $\nu = 0.3$, and restitution coefficient $e_n = 0.8$. The diameter ($2R = 4$ mm), density ($\rho = 2500$ kg/m^3), friction coefficient ($\mu = 0.6$), and size ratio ($\mathscr{R} = 7$) are identical to those for the analytic solution. For this problem, assuming a characteristic velocity $V_0 \sim \sqrt{2gR} = 0.198$ m/s, the normal spring stiffness for the linear spring model is selected to match the Hertzian contact duration [44] using

$$k_n = 1.2024 \left(m^{*1/2} E^{*2} R^* V_0 \right)^{2/5}, \tag{1}$$

where m^* is the effective mass

$$\frac{1}{m^*} = \frac{1}{m_l} + \frac{1}{m_f}, \tag{2}$$

R^* is the effective radius

$$\frac{1}{R^*} = \frac{1}{R} + \frac{1}{r}, \tag{3}$$

E^* is the effective Young's modulus

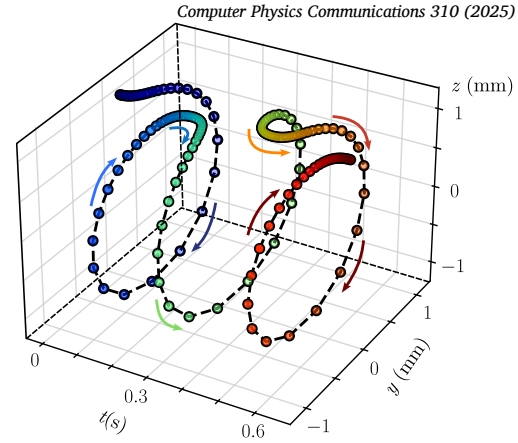

Fig. 3. Temporal evolution of fine particle position in the $x = 0$ plane for the conditions in Fig. 1 with $\theta_0 = 1°$, $\mathscr{R} = 7$, and $\mu = 0.6$.

$$\frac{1}{E^*} = \frac{(1-\nu_l^2)}{E_l} + \frac{(1-\nu_f^2)}{E_f}, \tag{4}$$

and the subscripts l and f denote large and fine particles, respectively. For these parameters values, $k_n = 3.26{\times}10^5$ N/m. The tangential spring stiffness is calculated using the approach suggested by Johnson [45]:

$$\frac{k_t}{k_n} = \frac{(1-\nu_l)/G_l + (1-\nu_f)/G_f}{(1-\nu_l/2)/G_l + (1-\nu_f/2)/G_f}, \tag{5}$$

where G is the shear modulus. For our case, this reduces to [44]:

$$\frac{k_t}{k_n} = \frac{2(1-\nu)}{2-\nu}. \tag{6}$$

For $\nu = 0.3$, $k_t = 0.8235 k_n = 2.68{\times}10^5$ N/m. To calculate the appropriate timestep, we first estimate the contact duration for a linear spring model:

$$t_c = \pi \sqrt{\frac{m^*}{k_n}}. \tag{7}$$

For the particle properties in this example, $t_c \approx 2.7$ µs. Note that this applies to elastic contacts, which have shorter durations compared to inelastic contacts, making the elastic case the most restrictive. To ensure accurate discretization of the contact, the computational timestep should typically be less than $t_c/40$ [46,47]. In this case, however, we choose a significantly smaller timestep of $t_c/270 = 10$ ns to guarantee more precise discretization. We do not simulate rolling or twisting friction in order to isolate the issue associated with the static friction term arising from the velocity-Verlet integration scheme.

Time series of θ and ω for three values of θ_0 from simulation are plotted as dotted curves in Fig. 2. Although the analytical and simulation results align well up to the point where the fine particle separates from the two large particles in the analytic solution (for $\theta \le 61°$ and $\omega \le 425$ rad/s), the DEM simulation results continue after the fine particle should separate from the large particles, producing unphysical results that persist for the duration of the simulation. In other words, the DEM predictions indicate that after the fine particle starts at θ_0 with $\omega = 0$, it rolls down the valley between the large particles (increasing θ) with a corresponding increase in angular velocity ω, but it never separates from the large particles. Instead, the fine particle continues to $\theta = 180°$, where ω reaches its maximum. Subsequently, ω decreases to zero as θ reaches $-\theta_0$. The particle then reverses direction, rolling back to $\theta = \theta_0$, during which ω reaches a negative peak before returning to zero. This cycle repeats indefinitely, resulting in an unphysical pendular trajectory for the fine particle. Fig. 3 shows this unphysical trajectory in the $x = 0$ plane versus time using the coordinate system defined in Fig. 1, including the reversals in direction at $-\theta_0$ and subsequently at θ_0 after the initiation of the motion (see video v1 in supplementary materials). Fig. 4 shows the change $\Delta r_\oplus$ in radial distance $r_\oplus$ (shown in

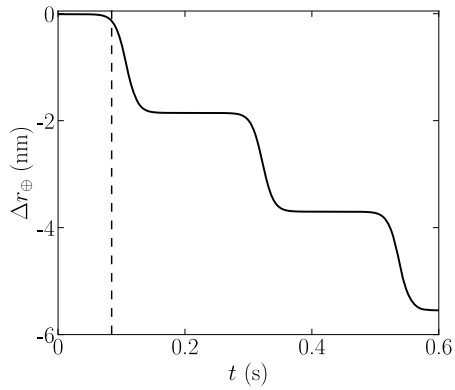

Fig. 4. Deviation, $\Delta r_\oplus$, of the radial distance $r_\oplus$ of the fine particle from the line connecting the centers of the two large particles calculated using DEM relative to that for hard spherical particles for $\theta_0 = 1°$, $\mathcal{R} = 7$, and $\mu = 0.6$. The vertical dashed line marks the time when, according to the analytical solution, the fine particle should separate from the two large particles. The negative sign indicates that the fine particle is moving deeper into the valley between the two large particles with each pendular oscillation.

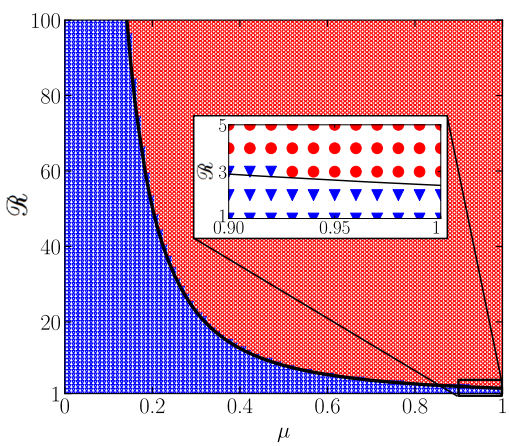

Fig. 5. Phase diagram showing when the fine particle separates (blue) or unphysically remains in contact (red) with the two large particles vs. size ratio, $\mathcal{R}$, and friction coefficient, μ, in DEM simulations for $\theta_0 = 5°$. 10,100 data points are plotted on a grid with a spacing of 0.01 in μ and 1 in $\mathcal{R}$. Black curve indicates the analytical expression for $\mathcal{R}_c(\mu)$, Eq. (22). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Fig. 1), of the fine particle compared to that for hard spherical particles, measured from the axis connecting the centers of the large particles to the center of the fine particle, during these oscillations. It starts with an initial negative value due to the initial overlap resulting from the 'soft' DEM particles; however, the subsequent increase in the negative value indicates that with each pendular oscillation, the fine particle unphysically moves deeper into the valley between the large particles, rather than separating from them.

To more systematically investigate the effects of size ratio and friction coefficient, 10,100 DEM simulations were conducted following the same approach. The values of $\mathcal{R}$ were varied in increments of 1 and μ in increments of 0.01, tracking instances where the fine particle followed an unphysical trajectory and where it separates from the two large particles. As shown in Fig. 5, unphysical behavior is not confined to extreme size ratios but can occur for size ratios as small as $\mathcal{R} = 3$ when μ is large (see inset). Because most prior DEM studies focused on monodisperse particles or small size ratios as well as moderate friction coefficients, this unphysical behavior was unlikely to significantly affect the DEM simulations results. It is only as we consider wider ranges of particle sizes (large $\mathcal{R}$) that the unphysical results become evident.

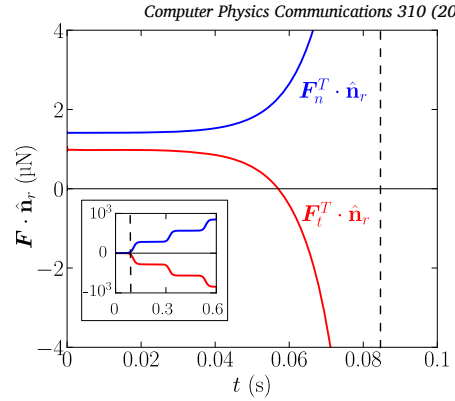

Fig. 6. Radial components of contact forces acting on the fine particle calculated using DEM for $\theta_0 = 1°$, $\mathcal{R} = 7$, and $\mu = 0.6$. $\boldsymbol{F}_n^T$ is the total normal force, and $\boldsymbol{F}_t^T$ is the total tangential force. The vertical dashed line marks the time when, according to the analytical solution, contact should have broken. The inset shows that the contact forces continue to increase with each pendular oscillation and become several orders of magnitude larger than their initial values. For context, the weight of the fine particle is about 2.4 μN.

Unphysical behavior also occurs when rolling friction or a twisting force is included in modeling the problem shown in Fig. 1. For the spring-dashpot-slider (sds) rolling friction model [48] used in LAMMPS, the fine particle exhibits pendular motion, but the oscillation amplitude decreases over time. In contrast, with Marshall's twisting friction model [49], the fine particle rolls down and eventually stops. In neither case does the fine particle fall off.

It is clear that this behavior predicted by the `linear_history` model is incorrect. Similarly, the `mindlin` model in LAMMPS, the `hooke` and `hertz` models in LIGGGHTS, and the `LinearViscoelasticFrictionSpecies` model in MercuryDPM all produce identical incorrect results. A sensitivity analysis using LAMMPS was performed to investigate the influence of timestep size. In the current setup, using the LS model with $\mathcal{R} = 10$ and $\mu = 0.6$, the fine particle only separates from the pair of large particles when a very small timestep, less than $t_c/2000$, is used. When μ is increased to 0.9, this threshold decreases further to $t_c/20000$. However, using such extremely small timesteps substantially raises computational costs, and the μ-dependence introduces uncertainty into the results.

The unphysical kinematics illustrated in Figs. 2 to 5 are associated with anomalies in the contact forces. Fig. 6 shows the net radial components (along a line from the x-axis connecting the centers of the two large particles to the center of the fine particle, which is the $\hat{\boldsymbol{n}}_r$ direction in Fig. 1) of the normal, $\boldsymbol{F}_n$, and tangential, $\boldsymbol{F}_t$, contact forces acting on the fine particle as a function of time for the $\theta_0 = 1°$ DEM simulation. To distinguish the total force from the individual contact forces, the superscript T in $\boldsymbol{F}_n^T$ and $\boldsymbol{F}_t^T$ denotes the total normal and tangential forces (sum of the pairwise contact forces for both large particles) acting on the fine particle. The vertical dashed line marks the time when, according to the analytical solution, the fine particle should separate from the two large particles.

At the onset of contact between the fine particle and the large particle, $\boldsymbol{F}_n^T$ and $\boldsymbol{F}_t^T$ are zero, but both change very quickly to become positive (which is not evident in Fig. 6 as it occurs within the first 10 μs). This behavior is a result of the soft-sphere approach in DEM. When the fine particle is released between the two spheres at the initial angle θ_0, the overlap is zero. However, as the fine particle moves downward due to gravity and begins to overlap with the large particles, a normal force is induced between the fine particle and each of the large particles. Concurrently, as the fine particle slides inward while overlapping, a tangential force also arises. Consequently, both the normal and tangential forces initially act in the radially outward direction from the centers of the large particles (positive $\hat{\boldsymbol{n}}_r$ direction) to counteract the radial component of the particle weight. From this time onward, $\boldsymbol{F}_n^T$

and $\boldsymbol{F}_t^T$ should both decrease as the radial component of the fine particle's weight decreases and the inertia related to centripetal acceleration associated with its circular trajectory between the two large particles increases. What is critical, though, is that the DEM code instead calculates a sharp unphysical increase in the magnitude of these contact forces, driving $\boldsymbol{F}_n^T$ strongly outward and $\boldsymbol{F}_t^T$ strongly inward (see Fig. 6). The increasing tangential force, $\boldsymbol{F}_t^T$, which acts radially inward, dominates, pushing the fine particle deeper into the valley between the two large particles, resulting in the unphysical behavior evident in Fig. 4. As a result of this incorrect calculation of $\boldsymbol{F}_n^T$ and $\boldsymbol{F}_t^T$, the fine particle does not separate from the two large particles at $\theta = 61°$, corresponding to the vertical dashed line at $t = 0.84$ s in Fig. 6. In fact, $\boldsymbol{F}_n^T$ and $\boldsymbol{F}_t^T$, as calculated by the DEM code, continue to increase, eventually reaching magnitudes several orders greater than their initial values, as shown in the inset of Fig. 6. This is clearly unphysical, because $\boldsymbol{F}_n^T$ and $\boldsymbol{F}_t^T$ are forces of constraint determined by the particle weight and the initial position of the fine particle.

This erroneous behavior is especially puzzling, as the particle's inward radial motion should be countered by the tangential force $\boldsymbol{F}_t^T$, but, paradoxically, this force also acts radially inwards, exacerbating the error. Instead of the fine particle rolling out of the valley between the two large particles, it is pushed deeper into the valley. The source of this anomaly lies in the velocity-Verlet integration scheme as described in detail in the next section.

3. Standard velocity-Verlet integration scheme

A DEM simulation begins by initializing the positions and velocities of particles and then models their subsequent development under the influence of contact and body forces. Various integration schemes are employed to numerically calculate this motion, with velocity-Verlet (along with its mathematically equivalent variant, the leapfrog method) being particularly popular due to its balance of efficiency and second-order accuracy compared to simpler integrators such as the Euler method, which are only first-order accurate [50,51]. The velocity-Verlet scheme requires only a single force calculation per timestep, unlike Runge-Kutta methods, which significantly reduces computational cost [52]. Additionally, it has modest memory requirements, as it does not necessitate storing positions, velocities, and accelerations from previous timesteps, unlike Adams-Bashforth or higher-order back-difference methods [52]. Crucially, the velocity-Verlet method is symplectic, meaning it preserves the geometric structure of Hamiltonian systems, which is necessary for conserving energy and momentum in simulations involving elastic, frictionless collisions. This balance of computational cost and accuracy makes the velocity-Verlet approach especially suitable for DEM problems where long-term stability and physical realism are essential [18]. As a result, it has been incorporated into several open-source DEM packages including LAMMPS, LIGGGHTS, MercuryDPM, and GRANOO.

3.1. Standard DEM calculation

A flowchart of the DEM timestepping process using the velocity-Verlet method is shown in Fig. 7. Each timestep begins with the initial position $\boldsymbol{x}$ and velocity $\boldsymbol{v}$ of each spherical particle. If it is the first timestep, the acceleration $\boldsymbol{a}$ is initialized to $\mathbf{0}$, whereas in later timesteps $\boldsymbol{a}$ is calculated from the previous timestep. Half-step velocities are computed as

$$\boldsymbol{v}\left(t + \frac{1}{2}\Delta t\right) = \boldsymbol{v}(t) + \frac{\Delta t}{2}\boldsymbol{a}(t), \tag{8}$$

which are then used to determine new positions at $t + \Delta t$:

$$\boldsymbol{x}(t + \Delta t) = \boldsymbol{x}(t) + \boldsymbol{v}\left(t + \frac{1}{2}\Delta t\right)\Delta t. \tag{9}$$

Pairwise overlaps are computed as

$$\delta = r_i + r_j - |\boldsymbol{dx}|, \tag{10}$$

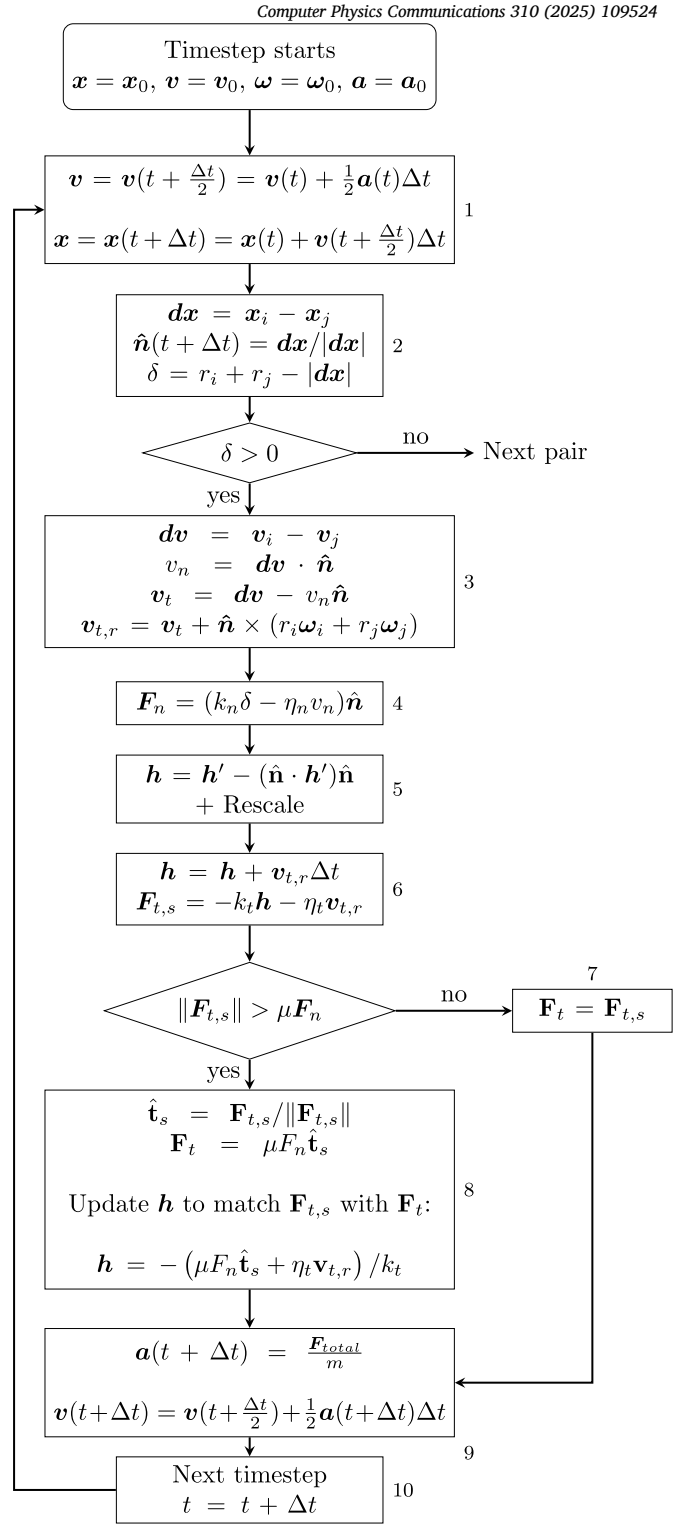

Fig. 7. Flowchart for DEM with a linear spring contact model using standard velocity-Verlet approach.

where r_i and r_j are the radii of particles i and j, and $\boldsymbol{dx} = \boldsymbol{x}_i - \boldsymbol{x}_j$. If the pairwise overlap is positive, the particles are considered to be in contact, and the pairwise contact force is calculated using a contact model. The Linear Spring (LS) model and the Hertz-Mindlin model are the two most commonly used contact models for simulating granular contacts. Here, we utilize the LS model for its simplicity, though the discussion and findings are equally applicable to the Hertz-Mindlin model for which a similar analysis is provided in Appendix B.

Before calculating the normal and tangential force components using the LS model, the relative velocity components between the contacting particles must be determined. The normal component is given by

$$v_n = \boldsymbol{dv} \cdot \hat{\boldsymbol{n}}, \tag{11}$$

where $\boldsymbol{dv} = \boldsymbol{v}_i - \boldsymbol{v}_j$ and $\hat{\boldsymbol{n}} = \boldsymbol{dx}/|\boldsymbol{dx}|$, where the caret indicates a unit vector. The relative tangential velocity at the contact point is calculated by subtracting the normal component from $\boldsymbol{dv}$ and adding the tangential component induced by the angular velocity $\boldsymbol{\omega}$ of the contacting particles:

$$\boldsymbol{v}_{t,r} = \boldsymbol{dv} - v_n\hat{\boldsymbol{n}} + \hat{\boldsymbol{n}} \times (r_i\boldsymbol{\omega}_i + r_j\boldsymbol{\omega}_j), \tag{12}$$

where the first two terms represent the center of mass translational tangential velocity, $\boldsymbol{v}_t = \boldsymbol{dv} - v_n\hat{\boldsymbol{n}}$. The normal force between contacting particles, $\boldsymbol{F}_n$, is then determined using the LS model and calculated as

$$\boldsymbol{F}_n = (k_n\delta - \eta_n v_n)\hat{\boldsymbol{n}}, \tag{13}$$

where k_n is the normal spring stiffness and η_n is the normal damping coefficient, which depends on the restitution coefficient (details are discussed in Section 5.1).

For tangential forces, the static friction component is modeled using a tangential spring. The challenge lies in implementing a Hookean linear spring for the tangential force, reflecting static contact before sliding occurs. By definition, a spring model requires a non-zero deformation during the "static" friction phase. This deformation is based on the tangential contact history from the time when contact between two particles begins at t_0 to time t within the current contact:

$$\boldsymbol{h} = \int_{t_0}^{t} \mathbf{v}_{t,r}(\tau)\,\mathrm{d}\tau, \tag{14}$$

where $\mathbf{v}_{t,r}$ is the relative tangential velocity at the contact point as defined in Eq. (12). This represents the elongation of the tangential spring, which might be thought of physically as shear deformation of the contacting particles relative to the point of contact before any sliding occurs.

In Step 6 of the flow chart in Fig. 7, the integral in Eq. (14) is represented as a summation, where the value of $\boldsymbol{h}$ is updated by adding the incremental tangential displacement, $\mathbf{v}_{t,r}\Delta t$, at each timestep. As the local coordinate system for the contact evolves from $\boldsymbol{x}(t)$ to $\boldsymbol{x}(t+\Delta t)$, the tangential displacement history $\boldsymbol{h}$, before its update in Step 6, is rotated to account for this rotation of the particle pair reference frame [48]. The typical approach to rotate the accumulated displacement into the tangent plane removes any component of the net tangential displacement that lies along the current normal direction, $\hat{\mathbf{n}}$, and rescales $\boldsymbol{h}$ to preserve its magnitude:

$$\boldsymbol{h} = \left(\boldsymbol{h}' - (\hat{\mathbf{n}} \cdot \boldsymbol{h}')\hat{\mathbf{n}}\right) \frac{|\boldsymbol{h}'|}{|\boldsymbol{h}' - (\hat{\mathbf{n}} \cdot \boldsymbol{h}')\hat{\mathbf{n}}|}. \tag{15}$$

This is shown in Step 5 where $\boldsymbol{h}'$ is the accumulated displacement prior to the current timestep and $\boldsymbol{h}$ is the corrected displacement.

The tangential static frictional force, is calculated based on the corrected displacement as:

$$\mathbf{F}_{t,s} = -k_t\boldsymbol{h} - \eta_t\mathbf{v}_{t,r}, \tag{16}$$

where η_t is the damping coefficient. As shown in the flowchart, when the magnitude of this tangential friction force exceeds Coulomb's sliding limit, $|\mathbf{F}_{t,s}| > \mu F_n$, where F_n is the magnitude of the normal force, the tangential force is recalculated using Amontons' first law, which assumes that the tangential force is proportional to the normal force with the dynamic friction coefficient μ as the proportionality constant:

$$\mathbf{F}_t = \mu F_n\hat{\mathbf{t}}_s, \tag{17}$$

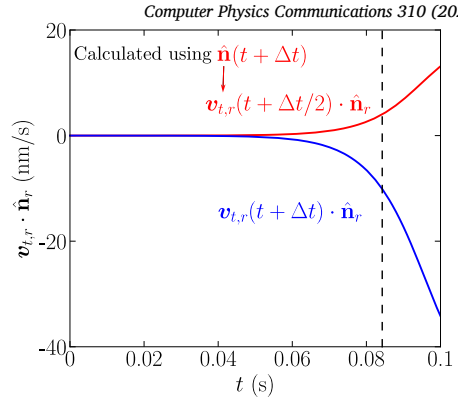

Fig. 8. Comparison of the radial components of incorrect half-step tangential velocity $\boldsymbol{v}_{t,r}(t+\Delta t/2)$ and full-step tangential velocity $\boldsymbol{v}_{t,r}(t+\Delta t)$ for $\theta_0 = 1°$, $\mathscr{R} = 7$, and $\mu = 0.6$. The vertical dashed line marks the time in the analytical solution when contact breaks.

where $\hat{\mathbf{t}}_s$ is the unit vector in the direction of $\mathbf{F}_{t,s}$. To account for the pure sliding state of the contact, along with variations in the normal force and relative tangential velocity, the tangential history $\boldsymbol{h}$ is updated to match the sliding tangential force [48]:

$$\boldsymbol{h} = -\frac{1}{k_t}\left(\mu F_n\hat{\mathbf{t}}_s + \eta_t\mathbf{v}_{t,r}\right). \tag{18}$$

This represents the maximum tangential spring elongation possible for an instantaneous value of normal force and relative tangential velocity. These pairwise normal and tangential forces, along with any other body forces, are then summed to get the total force acting on the particle, $\boldsymbol{F}_{\text{total}}$, which is then used to calculate acceleration at time $t+\Delta t$:

$$\boldsymbol{a}(t+\Delta t) = \frac{\boldsymbol{F}_{\text{total}}}{m}. \tag{19}$$

The acceleration is then used to update the half-step velocity to the full-step velocity, $\boldsymbol{v}(t+\Delta t)$, as

$$\boldsymbol{v}(t+\Delta t) = \boldsymbol{v}\left(t + \frac{\Delta t}{2}\right) + \frac{1}{2}\boldsymbol{a}(t+\Delta t)\Delta t. \tag{20}$$

Similar calculations update the torques and angular velocities of the particle pair. This is the standard velocity-Verlet integration scheme that is implemented in major open-source codes like LAMMPS, LIGGGHTS, and MercuryDPM at the time of publication.

3.2. Source of the error

In Figs. 4 and 6, we noted that the tangential force increases in the radially inward direction (toward the axis connecting the centers of the large particles) as the fine particle rolls down the valley between the two large particles, causing an inward motion that is physically unrealistic. From Step 6 of the flowchart in Fig. 7, the static tangential force depends on the history term $\boldsymbol{h}$, which is updated using Eq. (15). The history term relies on the relative tangential velocity $\boldsymbol{v}_{t,r}$ at the contact point, calculated in Step 3 using Eq. (12). In this calculation of $\boldsymbol{v}_{t,r}$, the normal vector $\hat{\boldsymbol{n}}(t+\Delta t)$ is derived from the particle positions at $t+\Delta t$ (Step 2), while the velocities used are from $t+\Delta t/2$ (Step 1). This causes the velocities to lag the positions by $\Delta t/2$. This lag leads to an inaccurate projection of the velocity in the tangential direction, subsequently yielding an erroneous history term and ultimately resulting in an incorrect tangential force.

We compare the radial components of $\boldsymbol{v}_{t,r}(t+\Delta t/2)$ and $\boldsymbol{v}_{t,r}(t+\Delta t)$ in Fig. 8. Here, $\boldsymbol{v}_{t,r}(t+\Delta t/2)$ is calculated using the standard velocity-Verlet approach and includes a lag of half a timestep, while $\boldsymbol{v}_{t,r}(t+\Delta t)$ is obtained using the positions from Step 1 and the velocities from Step 9, therefore both positions and velocities are at $t+\Delta t$. The half-step velocity components at $t+\Delta t/2$ are used to calculate the forces acting on the particles, while the full-step velocities at $t+\Delta t$ are used to update the

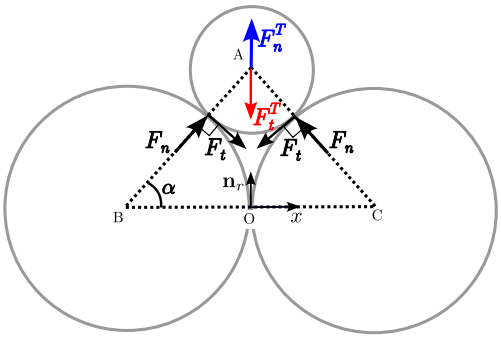

Fig. 9. Schematic of opposing net normal and tangential forces acting on the fine particle.

particle positions. As shown in Fig. 8, the incorrect half-step tangential velocity, $\boldsymbol{v}_{t,r}(t+\Delta t/2)$, acts radially outward and results in an incorrect tangential force acting radially inward. Since the history term $\boldsymbol{h}$ is the time integral of $\boldsymbol{v}_{t,r}(t+\Delta t/2)$, the error accumulates and $\boldsymbol{h}$ grows in the radially outward direction, resulting in a growing tangential force acting radially inward ($\boldsymbol{F}_t^T$ in Fig. 6). This incorrect tangential force is then used to calculate the full-step tangential velocity, $\boldsymbol{v}_{t,r}(t+\Delta t)$ in Step 9. Rather than being zero before the sliding limit and pointing radially outward after it, the full-step tangential velocity erroneously points radially inward, driving the fine particle deeper into the valley between the two large particles, ultimately preventing the fine particle from separating from them.

3.3. A simple physical model

After analyzing the error in the standard velocity-Verlet integration scheme, we now explain the influence of $\mathcal{R}$ and μ on the fine particle behavior, as shown in Fig. 5. The question is: why does the error in the velocity-Verlet scheme lead to unphysical results for certain combinations of $\mathcal{R}$ and μ, but not for others? The answer lies in the relative magnitudes of the net tangential contact force, F_t^T, and the net normal contact force, F_n^T, which act in the radial direction $\hat{\boldsymbol{n}}_r$, relative to the axis connecting the two large particles, as illustrated in Fig. 9. We assume that both large particles exert identical contact forces on the fine particle. The net normal force always acts outward from the valley, with a magnitude of $F_n^T = 2F_n \sin\alpha$, where α is defined in the figure and is determined by $\mathcal{R}$, i.e., $\cos\alpha = \mathcal{R}/(1+\mathcal{R})$ (see Eq. (35) in Appendix A). In contrast, the magnitude of the net frictional force is bounded by $F_{t,max}^T = 2\mu F_n \cos\alpha$ and can act in either the inward or outward direction. We are concerned here with cases where the net frictional force opposes the net normal force. For these cases, there are two distinct regimes depending on their relative magnitudes. In the absence of other forces, when $F_n^T > F_{t,max}^T$, the fine particle will always slip out of the valley between the two large particles; when $F_n^T < F_{t,max}^T$, the fine particle will be trapped. The boundary between these two regimes is determined by equating F_n^T and $F_{t,max}^T$, which gives

$$\mu = \tan\alpha. \tag{21}$$

When $\mu > \tan\alpha$, the fine particle will be trapped, and when $\mu < \tan\alpha$, the fine particle will slip out. Since α is determined by $\mathcal{R}$, an equivalent boundary between slipping and unphysical trapping can be expressed in terms of a critical size ratio, $\mathcal{R}_c$, and μ as

$$\mathcal{R}_c = \frac{1}{\sqrt{1+\mu^2}-1}. \tag{22}$$

Returning now to the results shown in Fig. 5, we see that the $(\mu, \mathcal{R})$ pairs associated with physical and unphysical behavior are well predicted by Eq. (22). This occurs because, although the standard velocity-Verlet integration scheme always unphysically increases F_t^T inward, it is only for $\mathcal{R} > \mathcal{R}_c$ that trapping can occur. When $\mathcal{R} < \mathcal{R}_c$, F_n^T is always larger than F_t^T, and the fine particle always slips out eventually because the radial component of gravity decreases and the centripetal acceleration increases as the fine rolls down the valley.

4. Improved velocity-Verlet integration

The unphysical pendular motion described above is ultimately due to the $\Delta t/2$ phase difference between the positions and velocities during force calculations in the standard velocity-Verlet algorithm. To address this issue, one might consider alternative integration schemes where positions and velocities are updated simultaneously. However, the velocity-Verlet algorithm offers several advantages that make it preferable over other methods, as described earlier. Moreover, the velocity-Verlet algorithm is already widely implemented in standard open-source codes, and switching to a different integration scheme could require significant modifications to the codebase. Therefore, in this section, we present an improved version of the velocity-Verlet algorithm that addresses the issue while retaining its inherent advantages.

In Steps 3 to 9 of the flowchart for the standard velocity-Verlet approach in Fig. 7, the primary goal is to compute the acceleration $\boldsymbol{a}(t+\Delta t)$ and use it to update the velocity $\boldsymbol{v}(t+\Delta t)$. This requires the forces and torques at time $t+\Delta t$. Using Taylor series expansion, the history term $\boldsymbol{h}(t+\Delta t)$ can be expressed as:

$$\boldsymbol{h}(t+\Delta t) = \boldsymbol{h}(t) + \left.\frac{d\boldsymbol{h}}{dt}\right|_t \Delta t. \tag{23}$$

A central difference expansion of the derivative

$$\left.\frac{d\boldsymbol{h}}{dt}\right|_t = \frac{\boldsymbol{h}(t+\Delta t/2) - \boldsymbol{h}(t-\Delta t/2)}{2(\Delta t/2)} \tag{24}$$

combined with the fact that the history term represents a displacement that can be updated using the velocity as

$$\boldsymbol{h}\left(t+\frac{\Delta t}{2}\right) = \boldsymbol{h}\left(t-\frac{\Delta t}{2}\right) + \boldsymbol{v}_{t,r}\left(t+\frac{\Delta t}{2}\right)\Delta t, \tag{25}$$

yields:

$$\boldsymbol{h}(t+\Delta t) = \boldsymbol{h}(t) + \boldsymbol{v}_{t,r}\left(t+\frac{\Delta t}{2}\right)\Delta t. \tag{26}$$

This approximation allows the relative tangential velocity at time $t+\Delta t/2$ to be accurately calculated. The modifications to the velocity-Verlet algorithm to accommodate this are shown as a flowchart in Fig. 10. In Step 3, before calculating the velocity components, we first compute the half-step normal vector $\hat{\boldsymbol{n}}'$ at $(t+\Delta t/2)$ using the updated position $\boldsymbol{x}(t+\Delta t)$ and velocity $\boldsymbol{v}(t+\Delta t/2)$. The half-step normal and tangential velocity components, $v_n(t+\Delta t/2)$ and $\boldsymbol{v}_{t,r}(t+\Delta t/2)$, are then determined using this $\hat{\boldsymbol{n}}'(t+\Delta t/2)$, rather than $\hat{\boldsymbol{n}}(t+\Delta t)$ in Step 2 of Fig. 7. Since both the normal vector and the velocities are evaluated at half-steps, there is no phase lag.

Using the modified approach, the half-step normal velocity $v_n(t+\Delta t/2)$ is then used to calculate the damping term for the normal force in Step 4. For the tangential force in Step 6, the history displacement term from the previous timestep is first rotated to the half-step plane using $\hat{\boldsymbol{n}}'$ to obtain the half-step history $\boldsymbol{h}_{1/2} = \boldsymbol{h}(t+\Delta t/2)$. This half-step history is then updated by adding the relative tangential velocity $\boldsymbol{v}_{t,r}(t+\Delta t/2)$, multiplied by timestep Δt. The resulting history term is then rotated from the half-step plane to the full-step plane using $\hat{\boldsymbol{n}}$ and rescaled. This updated history term is used to calculate the elastic component of the tangential force. The relative tangential velocity used in the damping term is also rotated and rescaled from the half-step plane to the full-step plane using $\hat{\boldsymbol{n}}$. These modifications ensure that the velocity components are in phase with the positions. The elastic normal force remains unaffected by these modifications, as it continues to use positions at $(t+\Delta t)$. The complete flowchart for the Hertz-Mindlin model with the improved velocity-Verlet algorithm is included in Appendix B.

The influence of these modifications on the forces is shown in Fig. 11, which corresponds to Figs. 4, 6 and 8 for the unphysical case. In addi-

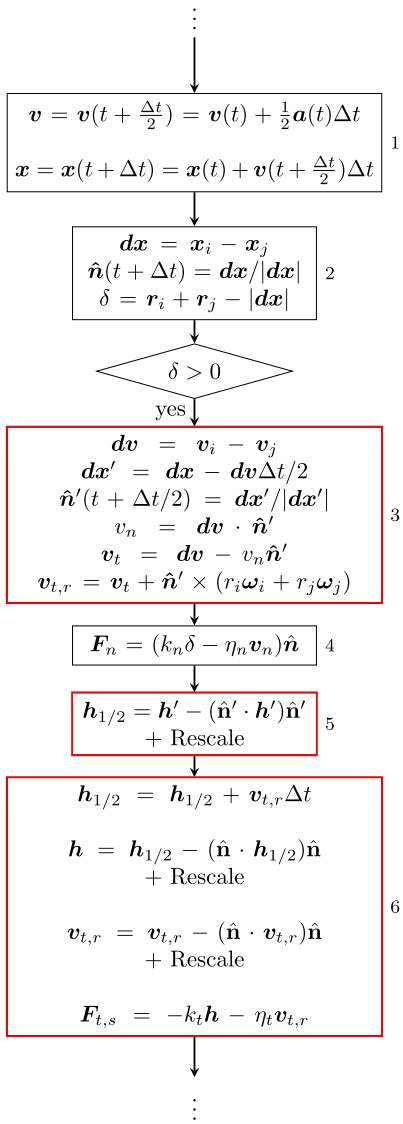

Fig. 10. Flowchart showing modifications to the velocity-Verlet algorithm.

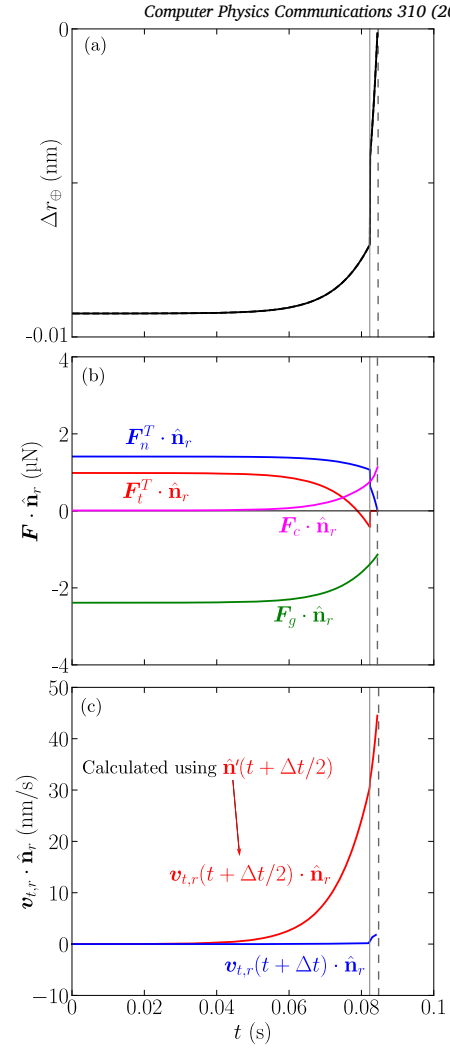

Fig. 11. DEM simulation results for geometry in Fig. 1 using improved velocity-Verlet integration scheme for $\theta_0 = 1°$, $\mathscr{R} = 7$, and $\mu = 0.6$. (a) Deviation of the radial distance of particle to the axis between the large particles from that for hard particles, (b) radial components of forces acting on the fine particle, and (c) radial components of the half-step and full-step tangential velocity. Vertical solid line indicates when the sliding limit is reached by tangential spring and the vertical dashed line indicates when contact breaks.

tion to the contact forces, the radial components of the gravitational force, $\boldsymbol{F}_g$, and the centrifugal force representing the inertia related to centripetal acceleration, $\boldsymbol{F}_c$, are also shown in Fig. 11(b). At the onset of contact between the fine particle and the two large particles, the radial components of both the normal and tangential forces are positive and balance the radial component of the particle weight, $\boldsymbol{F}_g$. Unlike the unphysical case simulation results shown in Fig. 6, $\boldsymbol{F}_n^T$ is now strictly decreasing. As the fine particle rolls down the valley between the two large particles, the centrifugal force increases as it speeds up and the radial component of the gravitational force decreases. Due to this, the radial component of the half-step tangential velocity $\boldsymbol{v}_{t,r}(t+\Delta t/2)$ increases as indicated in Fig. 11(c). This half-step tangential velocity is then used to calculate the elastic component of the tangential force $\boldsymbol{F}_t$. As the tangential spring associated with static friction stretches, the resulting tangential force opposes the particle's outward tangential motion, increasing in the radially inward direction. This maintains the circular motion of the fine particle about axis connecting the large particles, as indicated by the zero full-step radial component of the tangential velocity $\boldsymbol{v}_{t,r}(t+\Delta t)$. Once the static friction force exceeds the sliding limit (indicated by the vertical solid line), the tangential force transitions to that associated with dynamic friction. As the fine particle slips, both the normal and tangential forces quickly go to zero as the fine particle slides out of the valley losing contact with the large particles. This is also evident in the deviation of the radial position of the fine particle with respect to the hard particle limit, $\Delta r_\oplus$, shown in Fig. 11(a). Initially, $\Delta r_\oplus$ is negative, reflecting the small overlap of the DEM soft particles. But $\Delta r_\oplus$ becomes less negative as the force balance changes for $t > 0.05$ s. When the sliding limit is reached, $\Delta r_\oplus$ decreases quickly to zero as the fine particle separates from the large particles (see video v2 in supplementary materials).

To verify the accuracy of this revised approach, we compare the DEM predictions for θ and ω with the analytical solution, as shown in Fig. 12 ($\mathscr{R} = 7$, $\mu = 0.6$). A close match between the predictions from the new approach and the analytical solutions, particularly the fact that the fine particle loses contact with the two large particles at $\theta \approx 61°$, indicates that the revised approach provides accurate and physically realistic results. In addition, the improved velocity-Verlet method is used to simulate the 10,100 cases included in Fig. 5. In every case the fine particle falls off as it should.

The improved velocity-Verlet integration also produces physically correct results for the limited number of cases we considered when rolling friction or a twisting force is included in the simulation. How-

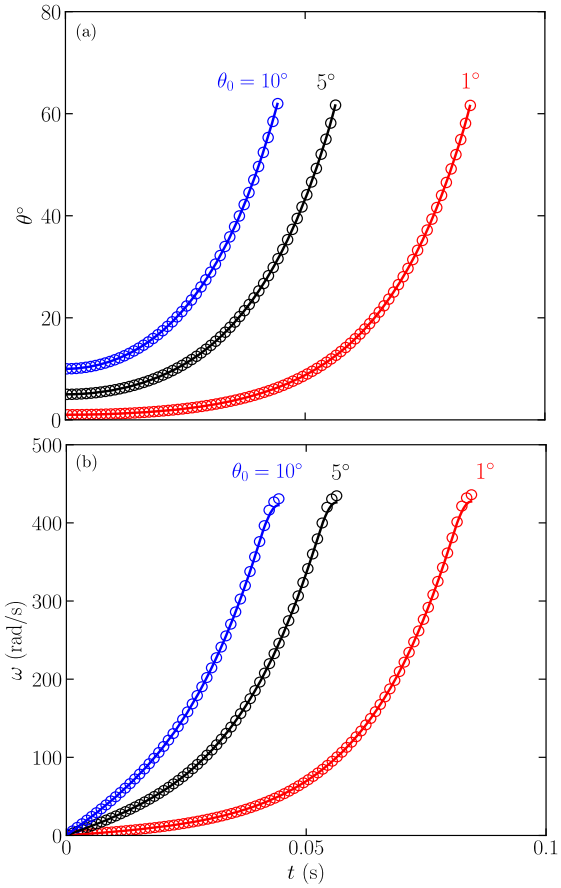

Fig. 12. Comparison of DEM predictions of θ and ω (circles) for a fine particle with analytical solutions (solid curves) and Fig. 1 geometry for different values of θ_0 using the improved velocity-Verlet algorithm ($\mathscr{R} = 7$, $\mu = 0.6$).

ever, a detailed investigation of these effects is beyond the scope of this work.

In Fig. 12, note that the angular velocity at the end of the contact predicted by DEM is slightly higher than the analytical values for all three values of θ_0. This difference arises because the analytical solution assumes a rigid sphere, whereas DEM simulations employ a soft sphere approach.[1] Additional tests for small size ratio two-body oblique collisions show that the improved velocity-Verlet approach results match those of the standard approach, indicating that the improved approach maintains accuracy for small size ratio two-body problems.

In methods like Dissipative Particle Dynamics (DPD) [53,54] and Smooth Particle Hydrodynamics (SPH) [55–60], adjustments are made to the velocity-Verlet algorithm to compensate for the time lag between particle positions and velocities [61,62]. While these adjustments are effective for those methods, the proposed improvement is crucial for DEM simulations to ensure that velocity components are computed without lag relative to position vectors because of the integral in the history term, which leads to error accumulation that the modifications to the DPD and SPH methods do not address. The proposed modification can also be combined with improvements to the DPD and SPH methods to further improve the accuracy.

[1] The soft sphere approach causes particle overlap, altering $r_\oplus$ and generating a radial tangential force.

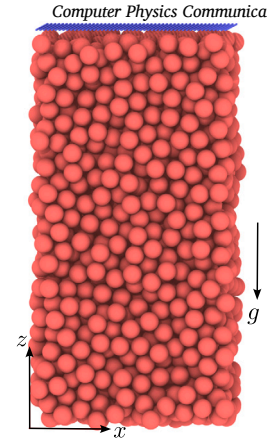

Fig. 13. A static bed of large particles (red) with $R = 2$ mm. Fine particles (blue), $\mathscr{R} = 20$, fall from just above the bed and percolate downward due to gravity. The density for both large and fine particles is 2500 kg/m^3, $e_n = 0.8$, and $\mu = 0.3$.

5. Fine particle percolation in static beds

To highlight the significance of using the improved approach in practical problems with large size ratios, we simulate the percolation of fine particles through a static bed of large particles. This example captures an ensemble of initial and contact conditions for fine-large particle interactions. In addition to implementing the improved velocity-Verlet algorithm, careful selection of DEM parameters, such as spring stiffness and timestep, is required for accurate simulations, particularly in granular systems with large size ratios. Thus, we begin by providing guidelines for selecting the appropriate DEM parameters, followed by a demonstration of the impact of using the revised integration scheme compared to the original one.

5.1. Guidelines for selecting parameters in large size ratio DEM simulations

For large size ratios, careful selection of DEM simulation parameters is critical, particularly to account for particle overlap. A 1% overlap, typical for similar-sized particles, translates to a problematic 10% overlap for fine particles at $\mathscr{R} = 10$. Consider first the normal spring stiffness, k_n, for a linear spring model. If k_n is too large, computational time increases due to smaller timesteps; if too small, excessive overlaps can lead to unphysical outcomes, such as fine particles embedding into larger ones. One approach bases the spring stiffness on material properties, matching the contact duration in a linear spring model to the Hertzian contact duration [44]. While effective for small size ratios dominated by normal contact forces, this method may not suffice for larger size ratios, where sliding and rolling effects become significant. We propose an alternative approach that determines k_n based on the maximum tolerable overlap. For a linear spring model, the normal spring stiffness k_n is related to the maximum overlap δ as

$$k_n = \frac{m^* V_0^2}{\delta^2}, \tag{27}$$

where m^* is the effective mass, V_0 is a characteristic relative velocity between particles, and δ is the particle overlap. For example, consider fine particles percolating through a static bed of large particles made of the same material, as shown in Fig. 13. Here, $V_0^2 \approx 2gR$, where g is the gravitational acceleration and R is the radius of the large particles. In this setup, the large particle diameter is $2R = 4$ mm, $\mathscr{R} = 20$, and the density of both large and fine particles is 2500 kg/m^3. To keep the overlap between large and fine particles below 0.1% of the fine particle diameter d, the minimum normal spring stiffness is 10,271 N/m using Eq. (27). The tangential spring stiffness, calculated using Eq. (5) (which simplifies to Eq. (6) for particles of the same material), is $k_t = 8458$ N/m for $\nu = 0.3$. The contact duration, determined using Eq. (7), is $t_c = 3.2$ μs,

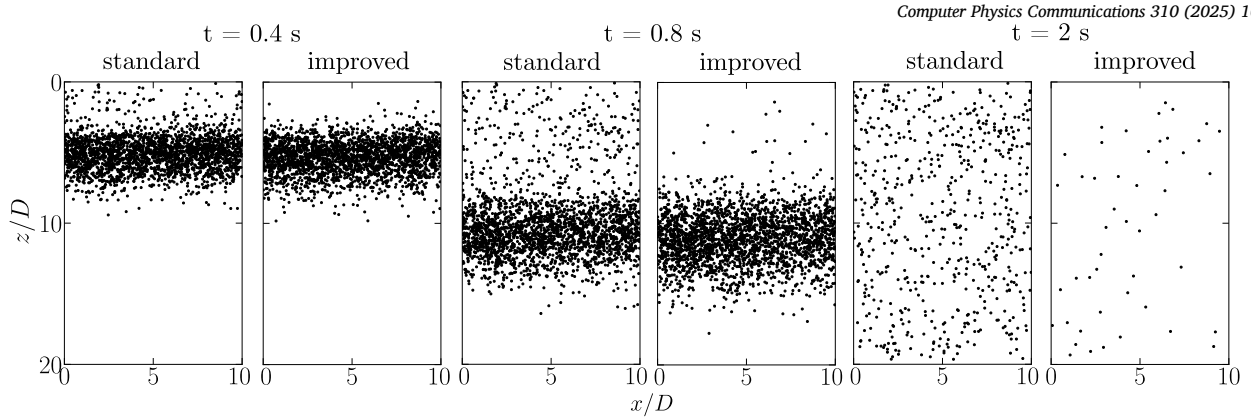

Fig. 14. Fine-particle positions at different times (t = 0.4, 0.8, 2 s) for $\mathcal{R} = 20$ and $\mu = 0.3$, comparing standard and improved velocity-Verlet algorithms. Note that trapped particles in the improved approach represent valid solutions of the soft particle contact model (see Eq. (21)) unlike most of the trapped particles in the standard case which do not.

which corresponds to a timestep of $t_c/40 = 0.08$ µs. A similar approach can be used for the Hertzian model based on the Young's modulus and Poisson's ratio [45]:

$$\delta = \left(\frac{15 m^* V_0^2}{16 R^{*1/2} E^*} \right)^{2/5}. \tag{28}$$

To ensure an overlap of 0.1% of the fine particle's diameter for our test case, the value of E^* is 2.2 GPa based on Eq. (28). For $\nu = 0.3$, this corresponds to $E = 4$ GPa. The tangential spring stiffness for the *mindlin* model in LAMMPS is calculated as $k_t = 8G^*$ [63], where G^* is the effective shear modulus:

$$\frac{1}{G^*} = \frac{2(2-\nu_l)(1+\nu_l)}{E_l} + \frac{2(2-\nu_f)(1+\nu_f)}{E_f}. \tag{29}$$

The elastic contact time can be approximated as [45]

$$t_c \approx \frac{2.94\delta}{V_0}. \tag{30}$$

For the case considered here, $t_c \approx 2.96$ µs and $t_c/40 = 0.074$ µs.

After selecting these parameters, the next step is choosing the appropriate damping model in LAMMPS. Traditionally, LAMMPS requires a normal damping coefficient, η_n, calculated from the desired coefficient of restitution, e_n, and a mean particle diameter. While this approach works well for small $\mathcal{R}$, it can produce inaccurate e_n values for high $\mathcal{R}$, as the calculated η_n does not account for particle size effects, leading to deviations from the specified e_n. To address this, we recently added a new damping model to LAMMPS, `coeff_restitution`, based on the method by Brilliantov et al. [64]. Also available in packages like LIGGGHTS [19] and MercuryDPM [21], this model requires only e_n as input. It calculates pairwise damping coefficients, accommodating both linear and Hertzian springs, effectively accounting for the effects of particle size and ensuring that the specified e_n is reproduced for each particle pair. Specifically, for a linear spring, the normal damping coefficient is:

$$\eta_n = \sqrt{\frac{4 m^* k_n}{1 + \left(\frac{\pi}{\log(e_n)}\right)^2}}, \tag{31}$$

and for a Hertzian spring, the normal damping coefficient is calculated at each timestep using

$$\eta_n = -2\sqrt{\frac{5}{6}} \frac{\log(e_n)}{\sqrt{\pi^2 + (\log(e_n))^2}} (R^* \delta)^{\frac{1}{4}} \sqrt{\frac{3}{2}\left(\frac{4}{3}E^*\right) m^*}. \tag{32}$$

LAMMPS also offers the `tsuji` model [65], which accepts e_n as input. However, two-sphere collision tests show that it does not accurately reproduce the specified e_n. The tangential damping coefficient η_t is scaled from the normal damping coefficient as $\eta_t = \zeta \eta_n$, where $\zeta = \sqrt{k_t/k_n}$ for the linear spring model and $\zeta = \sqrt{4G^*/E^*}$ for the Hertzian model [66].

Most DEM codes include models for rolling and twisting friction. LAMMPS provides the spring-dashpot-slider (`sds`) model, which uses the pseudo-force formulation by Luding [48], and the `marshall` twisting friction model [49], which is better suited for Hertzian contacts. The `marshall` model calculates the required twisting friction variables internally, eliminating the need for user-specified inputs.

5.2. *Effect of using improved velocity-Verlet algorithm*

To compare the standard velocity-Verlet algorithm and the new approach outlined in this paper, we follow the above guidelines for parameter selection and simulate the situation in Fig. 13, applying the linear spring model in both cases. The improved velocity-Verlet is enabled by using the `synchronized_verlet` flag in LAMMPS. A randomly packed bed of large particles, with an overlap of less than 0.1%, is prepared using a standard particle growth algorithm [67]. The monodisperse large particles have radius $R = 2$ mm, and the density of both large and fine particles is 2500 kg/m^3. The static bed of frozen large particles has dimensions of $20R \times 20R \times 40R$ and is periodic in the x and y directions, as shown in Fig. 13. Non-interacting fine particles with $\mathcal{R} = 20$ fall from 0.2 mm above the bed and percolate downward due to gravity. The fine particles interact with the bed particles but not with each other. The gravitational acceleration, g, is 9.81 m/s^2, the coefficient of restitution, e_n, is 0.8, and $\mu = 0.3$. The fundamental difference between this test case and the setup described in Fig. 1 is that here the fine particle is not initially in tangential contact with the large particles. Instead, it comes into contact with the bed of large particles at varying velocities and angles. This allows us to model a wide spectrum of initial conditions for a fine particle contacting one or more large particles.

We consider two cases where all parameters and other conditions are identical, including the static large-particle bed. The sole difference between them is the implementation of the velocity-Verlet algorithm. In the first case, we use the original version of the algorithm, referred to as the standard approach. In the second case, we employ the improved approach. The locations of 2500 non-interacting fine particles at three different times after their release are shown in Fig. 14, where the bed particles are made transparent to better visualize the positions of the fine particles.

When using the standard approach, many fine particles become unphysically trapped, as shown in Fig. 14, even though at $\mathcal{R} = 20$ fine particles should predominantly percolate freely through the bed because fine particles are much smaller than pore throats in the bed [68]. These trapped particles exhibit pendular motion similar to that shown in Fig. 3, preventing them from percolating through the bed. Some fine particles become trapped within $z/D < 10$ at $t = 0.8$ s. By $t = 2$ s, approximately

25% of the fine particles are trapped in the static bed. In contrast, with the improved velocity-Verlet approach, the majority of fine particles percolate freely, forming a dense band. This behavior aligns with the expected outcome for $\mathscr{R} = 20$.

A small fraction of fine particles, less than 2%, still become trapped even with the improved approach. However, unlike the erroneous behavior discussed in Section 2, this occurrence is consistent with the model's expected behavior for soft particle simulations as described in the context of Eq. (21). When $\mu > \tan\alpha$, fine particles can become trapped within the static bed of large particles when a fine particle with high velocity impacts and slides into a valley between large particles. In such cases, the large static friction generated during the initial contact prevents the fine particle from reaching the sliding limit, resulting in its entrapment. This is similar to a basketball getting stuck between the rim and the backboard. The softer the basketball, the more likely it gets stuck, although it still happens infrequently because the basketball needs to hit the rim and backboard in just the right way. In granular beds composed of large particles, the random packing and associated gaps between neighboring large particles further increases the likelihood of such physical trapping as they correspond to smaller α values at a given $\mathscr{R}$ compared to the case where large particles are touching. In the absence of twisting friction, these trapped particles exhibit pendular motion. However, unlike cases with the standard velocity-Verlet algorithm, they are not drawn deeper into the valley between large particles. While the pendular motion (in the absence of twisting friction) is physically unrealistic, the observed trapping is consistent with real-world physics for soft particles.

6. Summary

As computational techniques evolve and researchers apply DEM simulations are applied to new and complex domains, it is vital to proceed with caution, as demonstrated by the molecular dynamics code used in the supercooled water example [1] mentioned in the introduction. While expanding the application of DEM to more challenging problems is appealing, it also presents unforeseen challenges that can compromise the accuracy of simulations.[2] Our findings underscore the importance of rigorous validation and the need for careful consideration when extending established methods to novel research contexts and parameter regimes.

The improved velocity-Verlet integration scheme proposed in this paper addresses critical limitations in standard implementations of the DEM. We demonstrate that the conventional scheme can lead to unphysical results where fine particles remain in contact with larger particles due to the emergence of erroneous contact forces. This issue stems from the time discrepancy between velocity and position updates in the standard velocity-Verlet approach, which can result in the incorrect calculation of contact forces related to the friction history term for large size ratios, even at moderate levels of friction. The improved velocity-Verlet method resolves these inconsistencies by assuring that the velocity and unit normal vector are updated at the same time, rather than a half timestep apart. Since the computational cost of algebraic operations is orders of magnitude smaller than more expensive tasks like neighbor detection and writing output data, the additional computational overhead related to the improved velocity-Verlet algorithm is negligible.

While we have considered cases where fine particles are trapped in an unphysical pendular motion, both for a simple 3-particle system (Fig. 1) and a bed of particles (Fig. 13), the problem can occur whenever there are frictional contacts with large particle size ratios, with the exception of cases where friction is small (which is not typically the case for real particles in dense flows). Any dense flow with a relatively large size ratio could be affected by the problems related to the velocity-Verlet algorithm because the contacts are frictional and of relatively long duration. An ongoing detailed study of cases where these errors are significant suggests that with the same timestep size the improved velocity-Verlet approach produces more accurate results than the traditional method when overlaps are small. The proposed modification to the velocity-Verlet algorithm ensures accurate calculation of velocity components by synchronizing the evaluation of positions and velocities within the same timestep. Without this synchronization, the normal velocity component erroneously influences the calculation of tangential forces, leading to an inaccurate history term. This inaccuracy accumulates over time due to the integral nature of the history term associated with modeling static friction. Consequently, this improved velocity-Verlet algorithm should provide consistently more accurate results compared to the standard velocity-Verlet implementation at negligible computational cost.

Given the complexities associated with simulating systems involving large particle size ratios, we also offer guidelines for selecting appropriate spring stiffness, timesteps, and damping parameters for high size ratio simulations. These guidelines ensure the robustness and stability of simulations, especially when dealing with highly size-polydisperse systems. Simulations of fine particle percolation through a static bed of large particles demonstrate the usefulness of improved approaches in providing a more accurate and physically consistent description of particle behavior. With recent advances in computational power enabling the simulation of granular systems with increasingly wider ranges of size-polydispersity that capture the complexity of real-world industrial and geophysical granular flows, it is necessary that simulations with size ratios $\mathscr{R} \geq 3$ utilize the improved velocity-Verlet method to maintain accuracy and avoid artifacts that can compromise results.

CRediT authorship contribution statement

Dhairya R. Vyas: Writing – original draft, Methodology, Investigation, Conceptualization. **Julio M. Ottino:** Supervision, Project administration, Funding acquisition. **Richard M. Lueptow:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization, Funding acquisition. **Paul B. Umbanhowar:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of AI-assisted tools in the writing process

During the preparation of this work, the authors utilized ChatGPT for improving writing and proofing. After employing this tool/service, the authors reviewed and edited the content as necessary and assume full responsibility for the content of the published article.

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Declaration of competing interest

The authors have no competing interests to declare.

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[2] Platforms like the Open Network on DEM Simulations (ON-DEM) [69] are valuable for tracking progress and ensuring consistency across open-source codes in the field as they evolve.

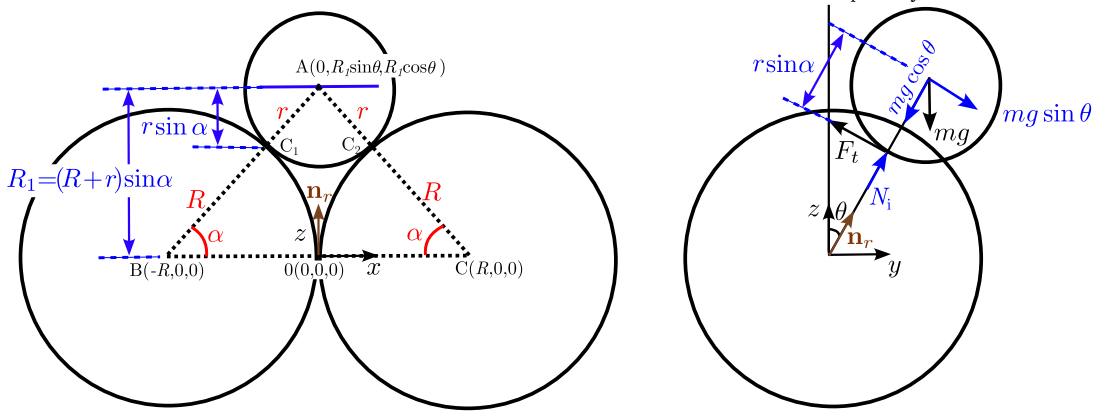

Fig. 15. Schematics showing (left) front view of the three particle geometry and (right) side view of forces on the fine particle.

Appendix A. Analytical solution

The geometry of the problem is shown in Fig. 15. Initially, and in the rigid body limit, the fine particle of mass m and radius r moves in a state of pure rolling. Once the tangential friction force F_t equals μF_n, the particle starts to slide in addition to rolling.

For initial pure rolling, the velocity of the fine particle, v_θ, can be determined by equating kinetic and gravitational potential energy. The gain in kinetic energy for an incremental change in the angle $\delta\theta$ of a particle undergoing pure rolling is

$$\Delta KE = \frac{1}{2}mv_{\theta+\delta\theta}^2 - \frac{1}{2}mv_\theta^2 + \frac{1}{2}I\omega_{\theta+\delta\theta}^2 - \frac{1}{2}I\omega_\theta^2, \tag{33}$$

where $I = 2mr^2/5$ is the moment of inertia of the fine particle and its the angular velocity is

$$\omega_\theta = \frac{v_\theta}{r\sin\alpha}, \tag{34}$$

where

$$\alpha = \cos^{-1}\left(\frac{R}{r+R}\right) = \cos^{-1}\left(\frac{\mathscr{R}}{1+\mathscr{R}}\right). \tag{35}$$

For $\delta\theta \to 0$, this simplifies to

$$\Delta KE = \left(\frac{5\sin^2\alpha + 2}{5\sin^2\alpha}\right) v_\theta m \frac{dv_\theta}{d\theta}\delta\theta. \tag{36}$$

For this incremental change $\delta\theta$, the change in potential energy

$$\Delta PE = mg(R+r)\sin\theta\sin\alpha\delta\theta. \tag{37}$$

Equating the ΔPE and ΔKE gives

$$v_\theta \frac{dv_\theta}{d\theta} = \frac{5\sin^2\alpha}{5\sin^2\alpha + 2} g(R+r)\sin\theta\sin\alpha. \tag{38}$$

Integrating Eq. (38) for the case where the velocity at the initial release angle θ_0, $v_\theta(\theta_0) = 0$, results in

$$v_\theta = (R+r)\sin\alpha\frac{d\theta}{dt} = \left[\frac{10g(R+r)\sin^3\alpha}{5\sin^2\alpha + 2}(\cos\theta_0 - \cos\theta)\right]^{1/2}. \tag{39}$$

The tangential force is

$$F_t = \frac{I\dot{\omega}}{r\sin\alpha}, \tag{40}$$

where the angular acceleration of the fine particle is

$$\dot{\omega} = \frac{d\omega}{dt} = \frac{1}{r\sin\alpha}\frac{dv_\theta}{dt}. \tag{41}$$

Note that both Equations (40) and (41) use the projected radius $r\sin\alpha$ instead of r. This is because, as shown in Fig. 15, the normal distance between the fine particle center and the contact points is $r\sin\alpha$. The total tangential force is evenly split between the two contact points C_1 and C_2 as

$$F_{t_1} = F_{t_2} = \frac{F_t}{2}. \tag{42}$$

The net normal force is

$$N = mg\cos\theta - m(R+r)\sin\alpha\left(\frac{d\theta}{dt}\right)^2, \tag{43}$$

which is also split between the two contacts as

$$N_1 = N_2 = \frac{N}{2\sin\alpha}. \tag{44}$$

When the fine particle begins to slide ($F_{t_i} > \mu N_i$), following Newton's second law, v_θ then evolves as

$$\frac{dv_\theta}{dt} = g\sin\theta - \frac{\mu N}{m\sin\alpha}. \tag{45}$$

Appendix B. Hertz-Mindlin model implementation

While the standard DEM calculation for the linear spring contact model was described in Section 3.1, here we provide a similar description for the Hertz-Mindlin model. The flowchart for the DEM steps involved in the Hertz-Mindlin with the improved velocity-Verlet algorithm model is shown in Fig. 16. Unlike the linear spring model, the Hertz-Mindlin model employs nonlinear Hertzian contact mechanics for normal forces and the Mindlin model for tangential forces in Step 4. The normal force is given by:

$$\boldsymbol{F}_n = (k_n' R^{*1/2}\delta^{3/2} - \eta_n v_n)\hat{\boldsymbol{n}}, \tag{46}$$

where the spring stiffness is $k_n' = 4E^*/3$, in accordance with Hertz theory. In the Mindlin model, the history-dependent no-slip solution differs from Eq. (16) of the linear spring model in Step 6 by including the radius of the contact area (patch), a_c, which evolves as $a_c = \sqrt{\delta R^*}$ as the contact progresses:

$$\mathbf{F}_{t,s} = -k_t' a_c \boldsymbol{h} - \eta_t \mathbf{v}_{t,r}, \tag{47}$$

where k_t' is spring stiffness per unit contact radius corresponding to the `mindlin` model. To match the Mindlin solution, one should set $k_t' = 8G^*$, where G^* is the effective shear modulus. When the tangential force exceeds the critical force defined by Coulomb's sliding limit, $\boldsymbol{h}$ is updated to match the sliding force as:

$$\boldsymbol{h} = -\frac{1}{k_t' a_c}\left(\mu F_n \hat{\mathbf{t}}_s + \eta_t \mathbf{v}_{t,r}\right) \tag{48}$$

in Step 8, similar to Eq. (18) for the linear spring model.

The improved velocity-Verlet approach is implemented for the Hertz-Mindlin model in a manner identical to the linear spring model. Specif-

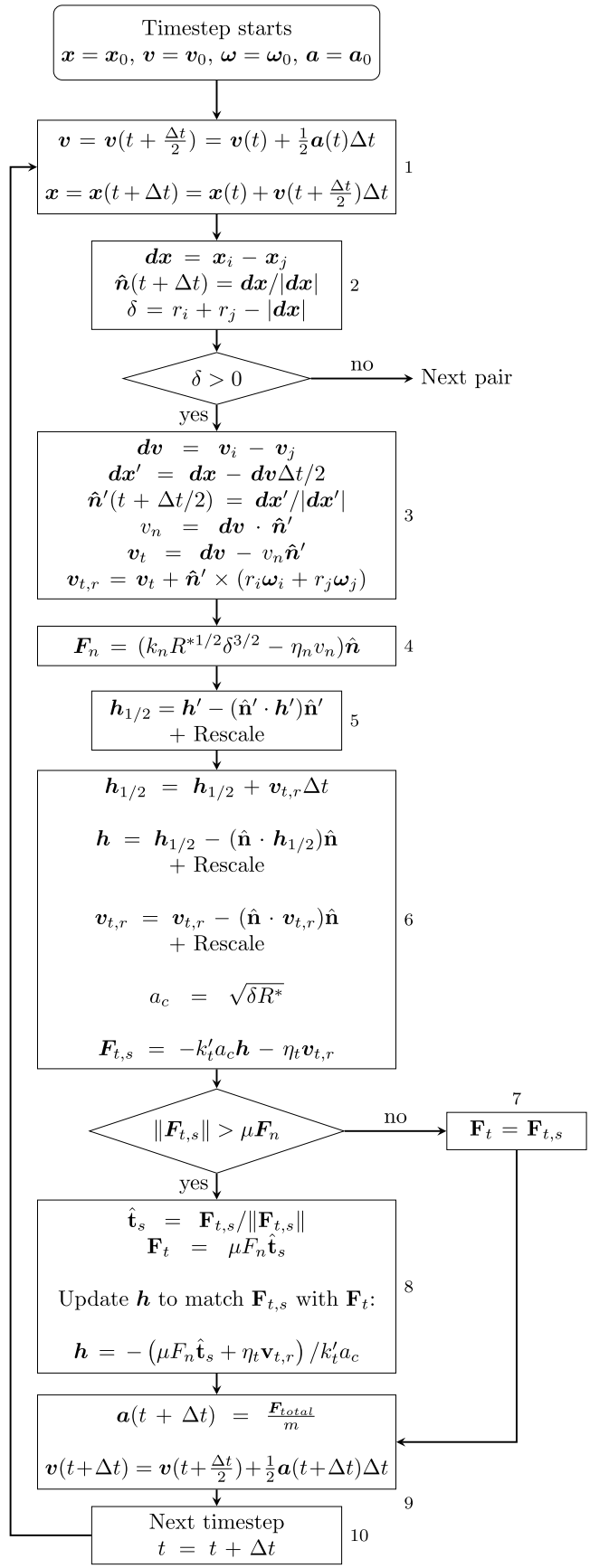

Fig. 16. Flowchart for DEM with Hertz-Mindlin contact model using improved velocity-Verlet approach.

ically, the history term is first rotated using the half-step normal vector $\hat{\boldsymbol{n}}'$, then updated, and finally rotated to align with the full-step normal vector (Steps 3, 5, and 6). This ensures that velocity and positions are synchronized, leading to accurate projections and forces.

Appendix C. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.cpc.2025.109524.

Data availability

The LAMMPS files for the improved velocity-Verlet integration scheme are available on Zenodo [70].

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